# Elkin Aguas

+33 (0) 7 67 48 37 40 | elkinaguas@gmail.com | elkinaguas | elkinaguas | elkinaguas

## Education

**Orange - Télécom SudParis** — *France*

Doctoral Research — *November 2018 - Present*

Theses topic: Automated Defense Systems for Cybersecurity

**IMT Atlantique (Former Télécom Bretagne)** — *France*

Master in Information Technologies — *September 2016 - September 2018*

Specialty: Communication network engineer

**District University of Bogota (Partner of IMT Atlantique)** — *Bogota, Colombia*

Bachelor Degree in Electronics (5 years) — *February 2010 - December 2015*

Specialty: Computer networks

## Publications

**On the Feasibility of Event-Driven Network Automation Scenarios for BGP** — *Paris, France*

**E. Aguas, T. Green, A. Lambert**

IEEE International Conference on Computer Communications (INFOCOM) — *2019*

**Towards a New Generation of Passive Optical Networks** — *Bogota, Colombia*

**E. Aguas, G. Puerto, C. Suarez**

Revista Ingeniería Universidad Distrital Francisco José de Caldas — *2016*

**Failure Modeling in a WDM Optical Burst Switched Network** — *Bogota, Colombia*

**E. Aguas, J. Calderón, G. Puerto**

2nd Workshop on Engineering Applications – International Congress on Engineering — *2015*

**Dynamic WDM-TDM Access Networks Featuring Wired-Wireless Convergence** — *Medellin, Colombia*

**E. Aguas, G. Puerto, C. Suárez**

Revista Facultad de Ingeniería Universidad de Antioquia — *2015*

## Experience

**Orange Labs - IP Networks, Monitoring and Security Department** — *Châtillon, France*

Research Internship - Event-Driven Network Automation Solution — *2018 (6 months)*

Design and implementation of an Event Driven Network Automation platform: State of the art, technological benchmark, implementation (salt, napalm, python, yang), writing of research articles. First scenario elaborated are about BGP hijacks automatic mitigation. They consist in automatically reconfiguring a network based on attacks detection. To avoid side effects due to automation, a machine learning component is being designed to enhance network self-healing.

**IMT Atlantique - Computer Sciences (INFO) Department** — *Brest, France*

Summer Internship - BGP, OSPF and SDN Simulation Development — *2017 (3 months)*

- Network design and implementation of BGP and OSPF protocols, and an SDN approach using Python and Quagga.
- Conception and implementation of BGP hijacking emulations, by using Python and Quagga, to help depict how the BGP protocol operates.

**District University of Bogota - GRECO Research Group** — *Bogota, Colombia*

Research Assistant - TDM-WDM Optical Networks and Wired-Wireless Convergence — *2014 (5 months)*

- Study the state of the art and propose a technology roadmap.
- Two articles were published from this project.

## Projects

**LWM2M Platform over an LPWAN Network - 5th semester project** *Rennes, France*

INTERNET OF THINGS *2017 (4 months) - Present*

Implement an LWM2M platform and define the compression rules to allow its use over an LPWAN network

**Visual Interface and Embedded Battery System for a Robot Coach - 4th semester project** *Brest, France*

ENGINEERING PROJECT *2017 (4 months)*

Provide guidance in the engineering and management aspects (project manager)

**Failure Detection and Analysis in an Optical Burst Switched Network** *Bogota, Colombia*

SOFTWARE ORIENTED PROJECT *2015 (6 mois)*

- Design and implementation in C++ of the optical network model used in the project and an algorithm for failure detection
- One article was published from this project

## Distinctions

June 2016 **Excellence scholarship**, from Telecom Bretagne for the program of master in France *Brest, France*

March 2017 **Meritorious thesis**, for the project "Failure Detection and Analysis in Optical Burst Switched Networks" *Bogota, Colombia*

## Skills

| | |
|---|---|
| Languages | Python, C++, JAVA, C, SQL, Matlab |
| Networks | TCP/IP, routing protocols (RIP, OSPF, BGP), optical networks (OBS), SDN security, mobile networks (4G, 3G, LTE) |
| Network orchestration | Salt (SaltStack), Ansible, Napalm, OpenStack |
| Computer Science | Computer networks, machine learning, scientific computing, databases, big data |
| Operating systems | Linux (Ubuntu), Linux (Debian), Windows |
| Software and tools | Spyder, Eclipse, Git, Quagga, Mininet, OMNeT++ |

## Languages

| | |
|---|---|
| Spanish | Mother tongue |
| English | Advanced (C1-TOEFL) |
| French | Intermediate level (B1) |

## Interests

Reading, Running, Biking, Photographie